

Framework for Probability Samples Learning Activity

STA304 — Summer 2026

Learning Activity 2 Overview

In this activity you will apply core probability and statistical concepts to construct and interpret inference from a probability sample. You will also compute the probability of selecting a sample or a unit under simple random sampling with and without replacement.

Exercises

1. Let $N = 6$ and $n = 3$. For purposes of studying sampling distributions, assume that all population values are known.

$$y_1 = 98 \quad y_2 = 102 \quad y_3 = 154$$

$$y_4 = 133 \quad y_5 = 190 \quad y_6 = 175$$

We are interested in $\bar{y}_{\mathcal{U}}$, the population mean. Two sampling plans are proposed.

- **Plan 1.** Eight possible samples may be chosen.

Sample Number	Sample, $\mathcal{S}$	$P(\mathcal{S})$
1	$\{1, 3, 5\}$	1/8
2	$\{1, 3, 6\}$	1/8
3	$\{1, 4, 5\}$	1/8
4	$\{1, 4, 6\}$	1/8
5	$\{2, 3, 5\}$	1/8
6	$\{2, 3, 6\}$	1/8
7	$\{2, 4, 5\}$	1/8
8	$\{2, 4, 6\}$	1/8

- **Plan 2.** Three possible samples may be chosen.

Sample Number	Sample, $\mathcal{S}$	$P(\mathcal{S})$
1	$\{1, 4, 6\}$	1/4
2	$\{2, 3, 6\}$	1/2
3	$\{1, 3, 5\}$	1/4

- (a) What is the value of $\bar{y}_{\mathcal{U}}$?
- (b) Let $\bar{y}_{\mathcal{S}}$ be the mean of the sample values. For each sampling plan, find:

$$(i) E[\bar{y}_{\mathcal{S}}]; \quad (ii) \text{Var}(\bar{y}_{\mathcal{S}}); \quad (iii) \text{Bias}(\bar{y}_{\mathcal{S}}); \quad (iv) \text{MSE}(\bar{y}_{\mathcal{S}})$$

- (c) Which sampling plan do you think is better? Why?
2. A public health researcher is selecting households for a wastewater survey. There are $N = 80$ households in the frame. The researcher uses a computer to generate an integer from 1 to 80 with equal probability. This is repeated until three unique households are identified; if the computer generates a number already in the set, that result is ignored, and the generator is run again.
- Is this procedure equivalent to Simple Random Sampling Without Replacement (SRS) or Simple Random Sampling With Replacement (SRSWR)? Justify your answer.
 - What is the probability of obtaining the specific unordered sample $\mathcal{S} = \{5, 18, 72\}$?
 - Calculate the inclusion probability π_j for household j .
3. A professor has a stack of 100 final exams. To ensure grading consistency, she decides to pick exams at random to double-check. Instead of setting aside the exams she has already checked, she uses a random number generator (1 to 100) to pick an exam, reviews it, and then returns it to the stack (meaning the same exam could be picked again). She repeats this process exactly 5 times.
- A student argues: "There are 100 exams and you picked 5 times, so the probability that my specific exam was checked is $5/100=0.05$." Is the student correct? Calculate the true probability. If you would like additional guidance, you may refer to the scaffolded steps provided below.
 - For any single draw, what is the probability that a specific exam (e.g. #42) is *not* selected?
 - What is the probability that a specific exam (e.g. #42) is *never* selected in all 5 independent draws?
 - Use this to calculate the probability that a specific exam is selected **at least once**.
 - What is the probability that the professor ends up reviewing ****exactly 4 unique exams****? (Hint: This means one exam was reviewed twice, and the others were all distinct). The scaffolded steps are provided below.
 - How many total sequences of exam IDs are possible (e.g., (1, 2, 3, 4, 5), (1, 1, 1, 1, 1), ...)?
 - If exactly 4 unique exams are chosen, one ID must appear twice and three IDs must appear once. How many ways can we choose which ID is the "double" and which three are the "singles"?
 - For a specific set of IDs like $\{A, A, B, C, D\}$, how many distinct ways can they be ordered in the 5 draws?
 - Combine these to find the final probability.

- c) If the professor wanted the probability of a specific exam being included to be *exactly* 0.05, should she have sampled **with** replacement or **without** replacement? Justify your answer mathematically.